8. Domain: Strategic Technologies

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	C-DAC has developed fully indigenous Bathymetric survey solutions for these high-altitude Himalayan glacier lakes. The Bathymetry survey solutions include three sub-Modules. i. Autonomous Bathymetric Survey Vessel (ABSV) ii. Advanced Glacier Lake Profiling Software (AGLPS) iii. Sensor Suite Module
	Name of the Project under which technology is developed	Autonomous Bathymetric Survey Vessel for Advanced Glacial Lake Profiling
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 8
	Brief Write-up on the Technology Developed	This project features an unmanned surface vessel (USV) designed to autonomously survey high-altitude glacial lakes, collecting high-resolution bathymetric data under extreme conditions. The vessel is equipped with: • Echo sounder for precise depth mapping and lakebed profiling. • An AI-powered obstacle avoidance system to navigate through complex environment. • ROS-based control system with autonomous waypoint navigation for efficient, remote operation. • Real-time telemetry and data logging to ensure reliable survey execution.
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